MSG No. 100.178 P_I4bm [Type IV, tetragonal]

Table 1: Wyckoff site: `4a`, site symmetry: `4..`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2,3,4]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[5,6,7,8]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,10,11,12]`
4	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[13,14,15,16]`

Table 2: Wyckoff site: `4b`, site symmetry: `2.mm`

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ z\right]$	`[1,4,7,8]`
2	$\left[0,\ \frac{1}{2},\ z\right]$	`[2,3,5,6]`
3	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,12,15,16]`
4	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[10,11,13,14]`

Table 3: Wyckoff site: `8c`, site symmetry: `..m`

No.	position	mapping
1	$\left[x,\ x+\frac{1}{2},\ z\right]$	`[1,8]`
2	$\left[\frac{1}{2}-x,\ x,\ z\right]$	`[2,5]`
3	$\left[x+\frac{1}{2},\ -x,\ z\right]$	`[3,6]`
4	$\left[-x,\ \frac{1}{2}-x,\ z\right]$	`[4,7]`
5	$\left[x+\frac{1}{2},\ x,\ z+\frac{1}{2}\right]$	`[9,16]`
6	$\left[-x,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[10,13]`
7	$\left[x,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	`[11,14]`
8	$\left[\frac{1}{2}-x,\ -x,\ z+\frac{1}{2}\right]$	`[12,15]`

Table 4: Wyckoff site: `16d`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$[-y,\ x,\ z]$	`[2]`
3	$[y,\ -x,\ z]$	`[3]`
4	$[-x,\ -y,\ z]$	`[4]`
5	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ z\right]$	`[5]`
6	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ z\right]$	`[6]`
7	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ z\right]$	`[7]`
8	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ z\right]$	`[8]`
9	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[9]`
10	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[10]`
11	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	`[11]`
12	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	`[12]`
13	$\left[-x,\ y,\ z+\frac{1}{2}\right]$	`[13]`

continued ...

Table 4

No.	position	mapping
14	$\left[x,\, -y,\, z+\frac{1}{2}\right]$	[14]
15	$\left[-y,\, -x,\, z+\frac{1}{2}\right]$	[15]
16	$\left[y,\, x,\, z+\frac{1}{2}\right]$	[16]